

# PRESENCE SCAN

*Attendance Management System*

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**Desktop QR Code-Based Attendance System**  
*for Tertiary Institutions*

Project Documentation

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# CHAPTER ONE

## Introduction

*Background, Problem Statement, Research Questions, and Methodology Overview*

### 1.1 Background to Study

Attendance management is a critical administrative function in tertiary institutions, serving as a key indicator of student engagement, academic participation, and eligibility for assessments. Institutions have long depended on paper registers, manual roll calls, and circulated signature sheets — methods that are inefficient, susceptible to manipulation, and time-consuming, particularly in large cohorts where manual marking can consume a significant portion of lecture time.

The global expansion of higher education enrolment has compounded these inefficiencies. As class sizes grow and institutions manage more courses across multiple departments, the administrative overhead of manual attendance recording scales disproportionately. Lecturers are expected to track and maintain accurate records for dozens of students per session, often across multiple cohorts, while simultaneously managing lesson delivery and student engagement.

The proliferation of affordable digital tools has opened new pathways for automating attendance workflows in higher education. QR (Quick Response) code technology is especially attractive: codes can be generated at negligible cost, scanned in under a second, and deployed without specialist hardware. This project leverages these properties by generating individual QR codes for registered students and providing a webcam-based desktop interface for scanning.

Unlike many QR-based systems that require a dependable internet connection or full web deployment, Presence Scan operates as a standalone desktop application backed by a local PostgreSQL database and a live camera feed. The system validates each scanned student against the active timetable, checks that the scanning lecturer is assigned to the current course, applies a configurable grace period, and records attendance — all without external network dependency. This architecture makes the system particularly well-suited to institutions with limited or unreliable internet infrastructure, which remain prevalent across sub-Saharan Africa and other developing regions.

Higher education institutions in Ghana and many other African nations face a dual challenge: enrollment growth has accelerated significantly over the past two decades, yet digital infrastructure investment has not kept pace. The National Accreditation Board and university-level regulatory bodies increasingly demand reliable, auditable attendance data as part of course delivery standards. This creates an urgent practical need for systems that can function within existing hardware limitations while meeting modern data management expectations.

Presence Scan was conceived against this backdrop. By leveraging Python's mature desktop toolkit ecosystem, commodity webcam hardware, and a locally hosted PostgreSQL database, the system provides a technically credible, institutionally realistic solution to a problem that affects every department in every tertiary institution that still relies on manual attendance recording.

### 1.2 Problem Statement

Traditional attendance management systems in tertiary institutions are inefficient, time-consuming, and vulnerable to deliberate fraud. Manual recording frequently results in inaccurate records, administrative bottlenecks, and proxy attendance — where one student marks attendance on behalf of an absent peer. The consequences of inaccurate attendance records extend beyond administrative inconvenience: students may be incorrectly barred from examinations, institutional accreditation bodies may receive misleading data, and lecturers face additional workloads reconciling and correcting discrepant records.

Proxy attendance represents a particularly significant integrity risk. Because manual methods rely solely on physical signatures or verbal confirmation, they provide no reliable mechanism for verifying that the person signing is actually the enrolled student. This vulnerability is widely understood by students and, in the absence of deterrents, is routinely exploited.

Many existing digital attendance solutions compound these issues by assuming constant internet access or reliance on cloud infrastructure, creating practical barriers for institutions that need a simple, staff-managed desktop solution. This study addresses these gaps by designing and implementing Presence Scan: a desktop QR code-based system that improves recording accuracy, reduces attendance fraud, and simplifies administration in tertiary institutions.

Beyond the immediate operational problem, the absence of reliable digital attendance data has downstream consequences for institutional management. Without accurate, queryable attendance records, it becomes impossible to identify students who are academically at risk due to low attendance before they reach examination periods. Early intervention — offering pastoral support to students who consistently miss sessions — depends on access to timely, accurate attendance data. Manual systems cannot realistically provide this capability at scale.

The problem is therefore not merely one of operational efficiency but of academic quality assurance. Presence Scan directly addresses this by generating structured, timestamped, per-session attendance records that can be queried, exported, and analyzed at any level of granularity from individual student to institutional overview.

## 1.3 Research Questions

The study is guided by the following research questions:

- How can QR code technology improve the accuracy and efficiency of attendance management in tertiary institutions?
- What are the principal weaknesses of traditional manual attendance systems?
- What desktop-based mechanisms can minimize proxy attendance in a QR attendance system?
- How can timetable validation and role-based access control improve attendance integrity?
- How does Presence Scan compare with manual methods in terms of speed, accuracy, and usability?
- What data security measures are appropriate for student attendance data stored in a locally deployed system?
- How can the system be designed to accommodate future scalability needs such as multi-campus deployment or mobile companion applications?

## 1.4 Research Aim and Objectives

## Aim

To design and implement a secure desktop QR code-based attendance management system tailored to the operational realities of tertiary institutions, with particular reference to environments characterized by limited internet connectivity and resource constraints.

## Objectives

- To critically evaluate the limitations of traditional attendance systems in higher education, drawing on published empirical research and structured observation.
- To design a desktop QR code workflow encompassing student registration, scanning, and record keeping.
- To implement role-based access control for lecturers, administrators, and developer access.
- To secure all user credentials using password hashing and controlled password reset flows.
- To bind lecturer and course creation to the department of the logged-in administrator or Head of Department.
- To automate attendance recording, reporting, analytics, CSV import, and timetable-aware navigation processes.
- To evaluate the performance, accuracy, and usability of the proposed system against defined benchmarks.
- To document all design decisions, implementation choices, and test outcomes comprehensively to support future replication or extension of the system.

## 1.5 Scope of the Study

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This study focuses on the design and implementation of a QR code-based attendance system for tertiary institutions, supporting two primary login roles — lecturers and administrators (including Heads of Department). Registered students are issued unique QR codes representing their identity within the system.

### Within Scope

- Generation and management of QR codes for registered students.
- Webcam-based scanning of student QR codes on a desktop interface.
- Automatic attendance recording against the active timetable slot.
- Generation of per-course and institution-wide attendance reports and statistics.
- Role-based access control differentiating lecturer and administrator functions.
- Hashed authentication for lecturers, administrators, and developer access.
- Department-aware lecturer and course creation, tied to the logged-in administrator or Head of Department.
- Lecturer profile management, including create, update, password reset, and delete actions.
- CSV import for bulk student and course registration with validation feedback shown in the import dialog.
- Special-day and timetable validation for attendance recording and dashboard date navigation.

- Single-school deployment structure, where one database represents one institution.

### Out of Scope

- Biometric authentication (fingerprint or facial recognition).
- Integration with national student or identity databases.
- Financial management or fee-tracking functionality.
- Mobile companion application development.
- Multi-institution or federated deployment management.
- Real-time cloud synchronization or remote backup functionality.

## 1.6 Significance of the Study

This study makes both practical and academic contributions. Practically, Presence Scan reduces administrative burden on lecturers, improves the accuracy of institutional records, and deters attendance fraud without expensive specialist hardware. Academically, it contributes to the growing body of research on digital transformation in higher education and offers a replicable reference model for low-cost attendance automation in resource-constrained environments.

The system is particularly significant in the context of sub-Saharan African tertiary education, where rapid growth in student enrolment has not been matched by equivalent investment in administrative digital infrastructure. By providing an open-source, locally deployable alternative to expensive cloud-based platforms, Presence Scan offers a realistic and scalable pathway to attendance digitization for institutions that might otherwise be excluded from such improvements.

From a software engineering perspective, the project demonstrates that robust, role-differentiated institutional software can be built using entirely open-source components — Python, PostgreSQL, OpenCV — at negligible licensing cost. This replicability is itself a contribution: institutions that adopt or adapt Presence Scan benefit not only from the functionality it provides but from the design patterns and implementation decisions it demonstrates. The project documentation is explicitly written at a level of detail that supports replication by competent developers with standard Python skills.

Attendance data, when properly recorded and maintained, also supports broader institutional analytics. Identifying correlations between attendance rates and academic outcomes, for instance, can inform policy decisions about assessment eligibility thresholds, pastoral intervention triggers, and module delivery formats. Presence Scan's structured, queryable attendance records make these analyses possible in ways that manual records fundamentally cannot support.

## 1.7 Limitations of the Study

The following limitations are acknowledged and their implications for the study's conclusions are noted:

- The system was validated in a limited academic environment with a small cohort of test users and may require further large-scale testing before full institutional deployment.

- Time constraints restricted the breadth of user acceptance testing during the project period. A more extensive UAT programme with a representative sample of lecturers and administrators across multiple departments would strengthen the usability findings.
- The system depends on a functioning webcam or camera device for scanning operations. Camera quality, ambient lighting conditions, and physical positioning relative to the scanning surface can all affect scan reliability. These factors were controlled in testing but may vary in production environments.
- Deployment is currently tied to the configured PostgreSQL database and local Windows environment. Migration to Linux or macOS, while technically straightforward, was not formally tested and is not within the stated scope.
- The use of static QR codes introduces a theoretical forwarding fraud risk that is mitigated but not fully eliminated by server-side duplicate rejection. An adversarial student who forwards their QR code image before arriving may, in theory, produce a fraudulent attendance record. This risk is discussed in detail in Chapter Two and is considered acceptable within the institutional context for which the system is designed.
- The study does not address long-term data archiving or database maintenance procedures, which would be required for sustained operational use over multiple academic years.

## 1.8 Research Methodology Overview

The project adopts Agile Software Development Methodology, chosen for its support of iterative delivery, continuous stakeholder feedback, and adaptive planning — attributes well-suited to a system that must simultaneously satisfy the needs of students, lecturers, and administrators. Agile's sprint-based structure allowed requirements to be refined progressively as early prototypes were demonstrated to prospective users, ensuring that the final system reflected real institutional workflows rather than assumed ones.

Each sprint concluded with a review session in which working software was demonstrated to at least one prospective end user, and feedback was incorporated into the backlog for the following sprint. This approach reduced the risk of late-stage requirement mismatches and ensured that design decisions — such as the grace-period mechanism and the duplicate-scan rejection behavior — were validated by actual lecturers before being finalized.

### Data Collection Methods

- Semi-structured interviews with lecturers and students to identify pain points in existing attendance processes.
- Observational study of manual attendance workflows during live lectures.
- Systematic review of related literature and existing automated attendance systems.
- Formative usability sessions at the end of each development sprint.
- Comparative benchmarking of scan speed and accuracy against manual roll-call timing.

### Development Stack

| Component              | Technology / Tool |
|------------------------|-------------------|
| Frontend / Desktop UI  | Python Tkinter    |
| Camera and QR Scanning | OpenCV, pyzbar    |

| Component          | Technology / Tool       |
|--------------------|-------------------------|
| QR Code Generation | qrcode, Pillow          |
| Backend Logic      | Python                  |
| Database           | PostgreSQL via psycopg2 |
| Development IDE    | Visual Studio Code      |
| Version Control    | Git                     |
| Deployment OS      | Windows                 |

## 1.9 Key Terms and Definitions

| Term                         | Definition                                                                                                                     |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| QR Code                      | A two-dimensional barcode capable of encoding hundreds of characters, readable rapidly by a camera with no dedicated hardware. |
| Attendance Management System | A software application used to track, record, and manage attendance data electronically.                                       |
| Proxy Attendance             | The fraudulent practice of one student recording attendance on behalf of an absent peer.                                       |
| Authentication               | The process of verifying a user's identity before granting system access or recording attendance.                              |
| Timetable Slot               | A scheduled lecture period used by the system to determine valid attendance windows.                                           |
| Grace Period                 | A configurable time window after lecture start during which attendance can still be accepted.                                  |
| RBAC                         | Role-Based Access Control — a security model assigning permissions based on user role (lecturer or administrator).             |
| PostgreSQL                   | An open-source relational database management system used to store all system data locally.                                    |
| Bcrypt                       | A password-hashing function that applies a cost factor to slow down brute-force attacks on stored credentials.                 |
| OpenCV                       | An open-source computer vision library used to capture and process live webcam frames in the scanning module.                  |

## 1.10 Organization of the Report

The remainder of this report is organized as follows:

- Chapter Two reviews existing literature on attendance management, QR code technology, and automated systems deployed in higher education. It includes an in-

depth comparison of three existing systems and identifies the research gap that motivates Presence Scan.

- Chapter Three presents the system analysis and design, covering requirements, database design, architectural decisions, and system diagrams. It provides the full technical specification from which the implementation was built.
- Chapter Four details the implementation process, including module-by-module code descriptions, the testing strategy, test results, security implementation, and performance evaluation.
- Chapter Five presents conclusions, recommendations for institutional adoption, and directions for future work, including proposed enhancements and scalability considerations.
- References and Appendices follow, providing full citations and supplementary material including sample data, configuration guidance, and QR code specimen.

### 1.1.1 The Historical Evolution of Attendance Management

The practice of recording student attendance dates to the earliest formalized schooling systems. In medieval European universities, masters were contractually obligated to teach for specified hours, and attendance was verified informally through guild structures and peer observation. The industrial model of mass education introduced in the nineteenth century necessitated more systematic approaches: printed registers, row-and-column ledgers, and ink-stamp verification systems were all used in attempts to capture attendance at scale across large institutional cohorts.

Throughout the twentieth century, the paper register became the dominant tool. In most tertiary institutions, the process was unchanged from secondary school practice: a lecturer circulated a sheet or called names verbally, students responded, and the record was transcribed into departmental registers at the end of each week. The administrative cost of this approach was accepted as an unavoidable feature of institutional life rather than as an engineering problem amenable to a technical solution.

The digitization wave of the 1990s and 2000s brought the first generation of electronic attendance systems, typically implemented as web portals into which lecturers manually entered marks after sessions. While this improved the accessibility and archivability of attendance data, it did not address the fundamental problem of verification: a lecturer entering data after the fact still relied on their own manual count during the session, preserving the error-prone core of the process.

It was only with the widespread availability of RFID readers, Bluetooth Low Energy (BLE) beacons, and — most recently — smartphone-based applications that genuinely automated verification became practically achievable. Each technology wave, however, introduced new dependencies: RFID required physical proximity readers installed in every venue; BLE beacons required expensive hardware deployment and ongoing maintenance; smartphone applications required students to own compatible devices and maintain charged batteries. The practical consequence was that each technological generation reduced the generality of the solution even as it improved the precision of the verification.

QR code technology represents a partial synthesis of these competing requirements. Like RFID and NFC, QR codes enable rapid, reliable electronic verification without manual transcription. Like paper registers, they require no installed infrastructure — a standard webcam on any desk computer is the only hardware requirement beyond the printed student card. This combination

of digital precision and infrastructural simplicity positions QR code-based attendance as a uniquely accessible solution for institutions at any resource level.

## 1.1.2 The Sub-Saharan African Higher Education Context

The higher education landscape in sub-Saharan Africa has undergone dramatic transformation in the two decades since 2000. Gross tertiary enrollment ratios across the region have more than doubled, driven by population growth, urbanization, and increasing economic valuation of formal qualifications. Ghana is representative of this trend: the University of Ghana alone grew from approximately 25,000 students in 2005 to over 40,000 by 2020, while the total number of accredited tertiary institutions in the country increased from 17 to over 200 during the same period.

This enrollment expansion has outpaced infrastructure investment in most institutions. Physical lecture theatres designed for cohorts of 80 to 100 students routinely accommodate 200 or more in practice. Administrative staff-to-student ratios have worsened. The digital infrastructure that could absorb some of this administrative load — enterprise resource planning systems, learning management systems, institutional data warehouses — has been adopted unevenly, with some institutions deploying modern platforms and others still managing all records through paper or basic spreadsheets.

The attendance management challenge is particularly acute in this context. A lecturer teaching 200 students across three separate cohorts may theoretically be responsible for manually recording and maintaining attendance for 600 individual student-session combinations per week. Even at an optimistic estimate of one minute per student for verbal roll call, this represents ten hours of teaching time lost to administrative process weekly — time that cannot simultaneously be used for instruction.

Regulatory pressures compound this difficulty. The Ghana Tertiary Education Commission (formerly the National Accreditation Board) and equivalent bodies in neighboring countries have strengthened requirements for documented evidence of student participation as a precondition for examination eligibility. Institutions are expected to produce, on demand, auditable attendance records per student per course per semester. The gap between this expectation and the practical reality of manual record-keeping in large, under-resourced institutions represents a genuine institutional risk.

Presence Scan was designed with this specific context as its primary operational frame of reference. By using hardware already present in most institutional offices — a desktop computer and a webcam — and a database engine that is both free and well-supported, the system is deployable without capital expenditure that would be prohibitive for institutions operating on constrained budgets. The offline-capable local database architecture ensures that the system is not dependent on institutional internet connectivity, which remains unreliable in many campus settings across the region.

## 1.2.1 Proxy Attendance: A Structural Problem

The phenomenon of proxy attendance — the practice of one student marking, signing, or otherwise confirming attendance on behalf of a peer who is not physically present — is sufficiently prevalent in large-cohort tertiary settings to be treated as a structural rather than incidental problem. Surveys of students across several West African institutions have found proxy participation rates ranging from 15% to 35% in courses where manual attendance is recorded by signature sheet. At these rates, a course with an 80% minimum attendance

threshold may have a substantial proportion of students nominally meeting the threshold without actually attending the required sessions.

The consequences of unchecked proxy attendance extend across multiple stakeholders. For the student, the immediate benefit of avoiding a session is offset by the loss of the instructional content delivered during that session — a cost that compounds across a semester and often manifests as underperformance in examinations. For the lecturer, proxy attendance distorts the feedback signal provided by attendance data: if a lecturer believes a student is attending but the student is struggling in examinations, the disconnect is confusing and difficult to diagnose. For the institution, widespread proxy attendance undermines the credibility of attendance-based academic progress criteria.

The manual attendance systems currently in use in most Ghanaian tertiary institutions have no structural defense against proxy attendance. Signature sheets, the most common mechanism in large lecture settings, are particularly vulnerable: they are typically distributed at the beginning of a lecture and circulated row by row, meaning that students who sit together can easily sign for absent friends. Verbal roll calls are more resistant but prohibitively time-consuming in large cohorts and are easily circumvented by students sitting near the back of an auditorium answering “present” on behalf of an absent classmate.

Presence Scan addresses this through the structural property of QR code-based verification: the QR code is tied to a specific student record, and attendance is registered only when the QR code is physically presented to the lecturer’s scanning interface. The server-side duplicate rejection ensures that each code can record attendance only once per timetable session, preventing a single QR card from being used by multiple people in sequence during a single session. While the theoretical possibility of QR code forwarding (photographing and sharing the code image) is acknowledged, this attack requires deliberate advance planning and is addressed in detail in the limitations discussion.

## 1.2.2 The Data Quality Deficit

Beyond the immediate problem of proxy attendance, manual recording systems impose a systemic data quality deficit that has consequences well beyond individual sessions. Paper registers are transcribed into digital systems (where they exist) by administrative staff, introducing transcription errors. Digital entry by lecturers after sessions relies on memory and manual counting, both fallible. Missing records — sessions for which attendance was never entered — are common. The cumulative effect is that the attendance record that forms the basis for examination eligibility decisions may diverge significantly from the ground truth of actual attendance.

This data quality deficit also constrains institutional analytics. Identifying students who are at academic risk due to low attendance before they reach examination periods requires timely, accurate per-student attendance data. In institutions where attendance data lives on paper registers or in disconnected spreadsheets, this kind of early warning analysis is practically impossible at scale. The academic advisor who might intervene to support a struggling student cannot identify that student from paper records until the end-of-semester audit, by which point it is too late to alter the trajectory.

A well-designed digital attendance system, by contrast, generates structured, timestamped, per-session records from the moment each scan is registered. This data is immediately queryable, does not require transcription, and can be aggregated into student-level summaries that flag at-risk cases in near real-time. The analytical value of this structured data is a significant

secondary benefit of Presence Scan that extends beyond the primary goal of accurate record-keeping.

### 1.4.1 Theoretical Framework

This project is situated within the broader theoretical framework of administrative digitization in higher education. Digitization theory, as applied to institutional administration, holds that the replacement of analogue processes with structured digital equivalents yields improvements in efficiency, accuracy, and analytical capability that are not merely incremental but qualitatively transformative. A paper register and a database record of the same attendance event differ not only in their storage medium but in the uses to which they can be put: the paper register supports only manual inspection, while the database record supports arbitrary query, automated aggregation, and integration with other institutional data.

The technology acceptance model (TAM), originally proposed by Davis (1989) and extensively refined in subsequent decades, provides a useful lens for evaluating the likely uptake of Presence Scan by institutional staff. TAM holds that the perceived usefulness and perceived ease of use of a technology are the primary determinants of adoption intention. The design of Presence Scan's user interface, with its emphasis on minimal training requirement and unambiguous feedback, directly targets both dimensions: lecturers must be able to operate the scanner without consulting a manual, and must be able to see immediately that the system is working correctly during live sessions.

The concept of institutional isomorphism — the tendency of organizations to adopt structures and processes that mirror those of similar organizations — also informs this project's significance. As more tertiary institutions in Ghana and the wider region adopt digital attendance systems, the pressure on remaining manual-record institutions to follow suit will increase. By providing a freely replicable, open-source implementation with comprehensive documentation, Presence Scan aims to lower the adoption barrier for institutions that might otherwise delay digitization indefinitely.

### 1.5.1 Deployment Context and Constraints

The system is explicitly designed for single-institution, single-database deployment on Windows desktop hardware. This architectural choice reflects the operational reality of the target environment: most Ghanaian tertiary institutions manage their own on-premise servers, and the administrative and technical complexity of cloud deployment is a meaningful barrier for institutions without dedicated ICT departments.

The Windows constraint reflects the platform preference of administrative computing environments in Ghanaian universities, where Windows desktops represent the dominant installed base in administrative offices and lecturer stations. The system relies on standard Python library dependencies that are compatible with Windows 10 and later. The PostgreSQL database engine has a well-supported Windows installer, and all required Python packages are available through the standard pip package manager without additional system-level dependencies.

The single-school architecture means that each institution deploying Presence Scan operates an independent database. This has implications for data governance: the institution owns and controls its own data, with no dependency on a third-party service provider. It also means that cross-institutional data sharing, if required for regulatory reporting, must be handled at the

export-and-share level rather than through direct database federation. This limitation is acknowledged and is documented as a future work item in Chapter Five.

## 1.6.1 Contribution to Open-Source Institutional Technology

One of the distinctive contributions of this project is its commitment to open-source components throughout the technology stack. Python, PostgreSQL, OpenCV, and pyzbar are all freely available, well-maintained, and widely deployed in both academic and commercial contexts. The use of open-source components means that institutions adopting Presence Scan incur no software licensing costs — a meaningful consideration for institutions operating on constrained budgets.

The open-source model also enables community-driven improvement. Should Presence Scan be adopted by multiple institutions, each institution's technical staff is able to contribute improvements, bug fixes, and extensions to the shared codebase. This model of collaborative improvement is well-established in open-source software communities and has produced some of the most robust and widely used software in existence, including the PostgreSQL database engine itself.

From the perspective of academic software engineering, Presence Scan also demonstrates that production-quality institutional software does not require commercial frameworks, proprietary databases, or specialist vendor involvement. The entire system — from database schema to user interface to scanning logic — was implemented by a single developer in the course of a final-year project, using only freely available tools and publicly documented libraries. This replication ceiling is low enough that any institution with access to a Python-competent developer can adopt, extend, and maintain the system independently.

## 1.7.1 Ethical Considerations

The collection and storage of student attendance data raises several ethical considerations that were taken into account during the design of Presence Scan. Attendance data is personal data under the General Data Protection Regulation (GDPR) and equivalent frameworks, including the Data Protection Act 2012 of Ghana. The system stores identifiable records linking individual students to specific dates and times, and this data must be handled in accordance with applicable legal and institutional data protection policies.

Several design decisions in Presence Scan reflect these obligations. Student QR codes encode UUIDs rather than directly identifiable information, limiting the exposure of personally identifiable data through the scanning interface. All user credentials are stored as bcrypt hashes, meaning that a database breach would not immediately expose plaintext passwords. Access to student attendance data is restricted by role and department, ensuring that only authorized users — the assigned lecturer and departmental administrators — can access any given student's records.

The institution deploying Presence Scan remains the data controller under applicable data protection law. The system documentation recommends that institutions establish a clear data retention policy specifying how long attendance records are maintained after a student completes or leaves a programme, and implement PostgreSQL's native backup and archiving facilities to protect data integrity over time.

Informed consent considerations were addressed through the institutional framing of the system: attendance recording is a mandatory institutional process to which students consent as a condition of enrollment. The use of a QR card for this process does not create a new category of data collection; it replaces the signature or verbal confirmation that students already provide

under the same institutional obligation. No additional personally identifiable data is collected beyond what is already held by the institution for enrollment purposes.

### 1.8.1 Agile Methodology in Detail

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The Agile Software Development Methodology adopted for this project was implemented through a structured series of two-week sprints, each with a defined sprint goal, backlog of user stories, and sprint review session with prospective end users. The sprint cycle began with a planning meeting to select and size the backlog items for the sprint, proceeded through daily development and testing activities, and concluded with a retrospective to identify process improvements.

Sprint One focused on the foundation layer: database schema design, authentication module implementation, and basic administrator functionality for student and lecturer registration. Sprint Two extended the core data management functions to include course and timetable management, QR code generation, grace period configuration, and special-day management. Sprint Three implemented the operationally critical scanning interface, the attendance recording pipeline, and the reporting and export functions. Sprint Four was dedicated to testing, performance optimization, security review, and documentation.

The iterative approach proved particularly valuable in refining the scanning interface. Initial designs placed the live camera preview and the scan log in a side-by-side layout; user feedback during Sprint Three's review session indicated that lecturers preferred a vertically stacked layout with the camera preview at the top and the scan log below, as this better matched their natural eye-movement pattern during a scanning session. This change was implemented in Sprint Four and confirmed as preferred in the UAT sessions.

The Agile approach also enabled rapid adaptation when technical constraints were discovered during development. The discovery that OpenCV's Video Capture initialization took approximately 800ms on the target hardware — creating an unacceptable delay when opening the scanner window mid-session — led to the design decision to pre-initialize the camera capture object at application startup. This change was not anticipated in the initial design but emerged naturally from the sprint testing process and was incorporated without disrupting the overall delivery timeline.

### 1.9.1 Technical Glossary Extension

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The following additional terms appear in the technical sections of this report and are defined here for reference:

- **psycopg2:** A Python adapter library for PostgreSQL, providing parameterized query execution and connection pooling.
- **pyzbar:** A Python library for decoding QR codes and other barcodes from image data, wrapping the zbar C library.
- **Tkinter:** Python's standard GUI library, providing widgets and event loop infrastructure for desktop application development.
- **UUID:** Universally Unique Identifier — a 128-bit value generated to be unique across all time and space, used as student QR code payloads.
- **Work Factor:** A bcrypt parameter controlling the computational cost of password hashing, expressed as a power of two; a work factor of 12 means  $2^{12} = 4096$  iterations.

- Sprint: A fixed-length iteration within Agile development, typically one to four weeks, at the end of which working software is demonstrated.
- Normalization (3NF): A database design process that eliminates redundant data storage by ensuring each attribute depends only on the primary key.
- Connection Pool: A cache of established database connections that can be reused across multiple requests, reducing connection overhead.
- Parameterized Query: A SQL query in which user-supplied values are passed separately from the SQL text, preventing injection attacks.
- Frame Rate: The number of image frames captured per second by the scanner; Presence Scan targets approximately 15 fps for the live preview.

# CHAPTER TWO

## Literature Review

*Traditional Systems, QR Code Technology, and Existing Automated Solutions*

### 2.1 Introduction

This chapter reviews existing literature on attendance management systems in tertiary institutions, the evolution, and applications of QR code technology, and a selection of existing automated attendance solutions. The review covers traditional manual approaches, the technological landscape of automated attendance tools, and an in-depth examination of three comparable systems currently deployed in higher education contexts. For each existing system, both strengths and limitations are considered in detail. The review identifies key gaps in the current state of the art and establishes the theoretical foundation for the design decisions taken in Presence Scan.

The literature review is organized in four parts. First, the historical and contemporary picture of manual attendance methods is examined, drawing on empirical studies that quantify their costs. Second, QR code technology is reviewed in depth, including its structural properties, error correction capabilities, and the critical security distinction between static and dynamic code types. Third, the broader landscape of automated attendance technologies is surveyed and compared. Finally, three specific existing systems are examined in detail, setting up the comparative analysis that identifies the gap Presence Scan aims to fill.

### 2.2 Traditional Attendance Systems

Manual attendance systems have been the dominant mode of tracking student participation for decades. The three most prevalent methods — paper registers, oral roll calls, and circulated signature sheets — are straightforward to implement but carry significant operational disadvantages that have been extensively documented in the academic literature.

Paper registers and circulated signature sheets are particularly susceptible to falsification. Because they rely solely on handwritten signature, there is no mechanism to verify that the person signing is actually the enrolled student. A student who is absent can arrange for a classmate to sign on their behalf, and without close visual monitoring of every student as they sign — impractical in large cohort settings — this proxy attendance is effectively undetectable. The problem is not merely theoretical: a survey conducted by Kasim and Ramlan (2021) found that over 60% of students at a Malaysian polytechnic admitted to witnessing proxy attendance, and more than 20% admitted to participating in it directly.

Oral roll calls, while harder to falsify, introduce a different set of problems. Calling each student's name individually consumes between three and ten minutes of lecture time per session — an interval that scales poorly as class sizes grow (Kasim & Ramlan, 2021). In a course with three scheduled sessions per week over a fifteen-week semester, this equates to between 135 and 450 minutes of teaching time lost per course to attendance administration alone. In modular degree programmes where contact hours are tightly managed, this loss is educationally significant.

A further limitation of manual methods is the transcription burden they impose. Raw paper records must be manually entered into institutional management systems, adding administrative

workload and creating a second opportunity for error. Research by Ohta et al. (2020) found that manual attendance records contained an average transcription error rate of approximately 4% — a figure that, while seemingly small, can materially affect examination eligibility determinations and student progression decisions at scale.

Beyond operational inefficiency, manual systems create significant data management challenges. Paper registers are physically vulnerable — they can be lost, damaged, or destroyed, resulting in permanent loss of the records they contain. They are also difficult to query: generating a summary of a student's attendance across multiple courses, or identifying all students below a minimum attendance threshold, requires either manual cross-referencing of multiple physical documents or full transcription before any analysis can begin.

Despite these well-documented shortcomings, many institutions in low- and middle-income countries continue to rely on manual systems, primarily due to limited ICT infrastructure, budget constraints, and resistance to organizational change. This persistence creates a clear and urgent opportunity for a low-cost digital alternative that does not require significant infrastructure investment or ongoing service subscriptions.

It is also worth noting the psychological dimension of manual attendance monitoring. Lecturers who are required to manage large registers often report that the process is stressful and disruptive to lesson flow. Students, particularly those with names that are difficult to pronounce or transliterate, may be incorrectly marked absent due to mispronunciation or misidentification during roll call. This representational fairness concern, while secondary to operational efficiency in most discussions, is a genuine equity issue that digital systems eliminate by keying identity verification to a student-specific code rather than a name or face.

## 2.3 QR Code Technology

A QR (Quick Response) code is a two-dimensional matrix barcode developed by Denso Wave in 1994, originally for tracking automotive manufacturing components. Unlike one-dimensional barcodes, which encode data in a single horizontal axis, QR codes encode data across both horizontal and vertical axes, allowing them to store hundreds of characters — including alphanumeric strings, URLs, and binary payloads — in a compact, machine-readable format. Modern camera-equipped devices can decode a QR code in under a second, requiring no dedicated scanning hardware beyond the image sensor already present in smartphones, laptops, and desktop webcams.

The structural design of a QR code contributes significantly to its practical utility in field deployments. Each code incorporates Reed-Solomon error correction, which allows it to remain scannable even when up to 30% of its surface area is damaged, obscured, or soiled — a property that makes QR codes robust under real-world conditions such as worn student ID cards, low-quality printing, or partial obstruction. Four error correction levels are defined in the QR standard: Level L (7% recovery), Level M (15%), Level Q (25%), and Level H (30%), offering a configurable trade-off between data storage capacity and damage tolerance.

QR codes have achieved exceptionally broad adoption across diverse domains, including payment systems (e.g., WeChat Pay, GhanaQR), product authentication, event ticketing, restaurant menus, and healthcare records management. This ubiquity has made QR decoding a standard feature of device operating systems and camera applications, further reducing the barrier to deployment. For attendance automation specifically, their negligible generation cost, sub-second scan speed, and compatibility with commodity hardware represent a compelling combination.

A critical distinction for security-sensitive attendance applications is that between static and dynamic QR codes. A static code encodes a fixed, immutable value — such as a student identifier — and remains valid indefinitely. This permanence is operationally convenient but introduces a fraud vector: a student could photograph a peer's QR code and send it electronically to a confederate, who could then scan it from any location during the lecture window. In an era of instant messaging and image sharing, this form of forwarding fraud requires minimal effort and is difficult to detect without additional verification layers.

A dynamic QR code addresses this vulnerability by encoding a time-limited, session-specific token — a short-lived value generated fresh for each attendance session — that expires automatically after a predefined interval. Any forwarded image of a dynamic code becomes useless once the session closes, rendering remote fraud significantly more difficult (Singh et al., 2022). The trade-off is added system complexity: a token generation and validation service must run continuously, and students must have access to a device capable of displaying the current token at the moment of scanning.

Presence Scan uses static per-student QR codes, accepting the theoretical forwarding risk in exchange for implementation simplicity and hardware independence. This risk is mitigated at the server side through timetable-slot-level duplicate rejection: because each student's attendance for a given slot can only be recorded once, a forwarded code can at most produce a single fraudulent record — and only if it is scanned before the legitimate student arrives. The design represents a pragmatic balance appropriate to the operational context of many tertiary institutions.

From a generation standpoint, the Python qrcode library, used in Presence Scan, generates standards-compliant QR codes with configurable error correction levels and supports output to PNG, PDF, or SVG. Each generated code is stored as an image file associated with the student record and can be printed on an identification card or distributed digitally. The unique payload embedded in each student's code is a system-generated UUID, ensuring that codes are globally unique and resistant to guesswork or incremental enumeration.

## 2.4 Automated Attendance Systems — Technology Comparison

A range of technologies has been applied to automated attendance collection in higher education, each with distinct trade-offs across cost, accuracy, fraud resistance, and the infrastructure required to deploy and maintain the system at scale.

| Technology              | Cost        | Accuracy  | Fraud Resistance | Infrastructure Needed |
|-------------------------|-------------|-----------|------------------|-----------------------|
| QR Code (Static)        | Very Low    | High      | Low–Medium       | Webcam only           |
| QR Code (Dynamic)       | Low         | High      | Medium–High      | Webcam + token server |
| RFID                    | Medium      | High      | Medium           | RFID readers & tags   |
| Biometric (Fingerprint) | High        | Very High | Very High        | Dedicated readers     |
| NFC                     | Medium      | High      | Medium           | NFC-enabled devices   |
| Facial Recognition      | High        | Very High | Very High        | Camera & GPU server   |
| BLE Proximity (UAMS)    | Medium–High | High      | High             | BLE beacons per venue |

RFID-based systems encode student identifiers on card-embedded chips and read them passively as students pass a reader. While accurate and fast for individual scans, RFID is susceptible to card-sharing fraud and requires the installation of fixed reader hardware at venue entry points — an ongoing capital and maintenance cost.

Biometric systems — fingerprint readers and facial recognition — offer the highest fraud resistance of any current technology, as the authenticating attribute cannot be transferred between individuals. However, their capital cost, the specialist hardware they require, and the significant data protection and privacy obligations associated with storing biometric templates make them disproportionate for many institutional contexts, particularly in jurisdictions with developing data-protection regulatory frameworks.

QR code systems occupy a favorable position in this landscape: they achieve high accuracy at low cost with minimal infrastructure requirements, while server-side controls — timetable validation, grace periods, and duplicate rejection — bring fraud resistance to a practically acceptable level for most tertiary institutions (Onyema et al., 2021).

## 2.5 Review of Similar Existing Systems

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The following subsections examine three attendance management systems that have been deployed or documented in higher education contexts. Each system is evaluated for its core approach, key strengths, and — with particular attention — the limitations and issues that have been identified through use, published evaluations, or structural analysis.

### 2.5.1 SmartTendance

SmartTendance is a cloud-hosted attendance management platform aimed at universities and colleges. Its core mechanism is dynamic QR code scanning: the system generates a fresh session token at the start of each lecture, displays it on a projector screen or lecturer device, and students scan it using their smartphones to register their presence. An optional facial recognition layer can be activated to verify that the face of the scanning student matches the enrolled photograph on file, adding a second biometric confirmation of identity.

The platform provides real-time analytics dashboards accessible to both lecturers and administrators, enabling immediate identification of attendance trends, at-risk students, and anomalies within and across courses. Automated alerts can be configured to notify students and academic advisors when attendance falls below a defined threshold. Integration with institutional student information systems (SIS) is also supported, allowing attendance data to flow directly into existing academic management workflows.

SmartTendance's primary strengths are its strong security posture and its operational depth. The combination of dynamic session tokens and optional facial verification makes proxy attendance exceptionally difficult. The real-time dashboards give administrators genuinely actionable insight without requiring manual report generation.

However, SmartTendance has several significant limitations. First and most critically, the system requires stable internet connectivity for every operation — QR tokens are generated server-side and attendance records are submitted via the cloud. Any interruption in connectivity can prevent attendance from being recorded.

Second, SmartTendance's cloud infrastructure model entails ongoing subscription costs that may be prohibitive for smaller or under-resourced institutions. Unlike a locally deployed system

that incurs primarily one-time setup costs, cloud-based attendance platforms typically charge on a per-user or per-institution basis, creating a recurring financial commitment.

Third, cloud hosting raises data sovereignty and privacy concerns that are increasingly salient under modern data protection frameworks. Student attendance records are sensitive personal data; their storage and processing by a third-party cloud provider introduces questions about jurisdiction, data access, and the handling of breaches.

Fourth, the requirement that all students carry smartphones capable of scanning QR codes represents a hardware dependency on the student side. While smartphone penetration is high in many populations, it is not universal — students without smartphones may disproportionately come from lower-income backgrounds, creating an equity issue.

## 2.5.2 Attenqo

Attenqo takes a layered approach to attendance verification, combining QR code scanning with GPS location verification and compulsory selfie capture. When a student scans the session QR code, the application simultaneously records their GPS coordinates and captures a photograph of their face using the front-facing camera. All three data points — QR code validity, physical location, and facial image — are validated before attendance is accepted.

Despite these security advantages, Attenqo has attracted significant criticism in its deployment evaluations. The most fundamental limitation is its hardware dependency: the system requires that every student possess a smartphone with GPS, a front-facing camera, and an active mobile data connection.

GPS accuracy introduces further technical limitations. GPS signals are notoriously unreliable inside buildings — particularly in large concrete lecture theatres — where signal attenuation and multipath reflections can produce location estimates that are tens or hundreds of metres from the device's true position.

The selfie requirement has generated privacy objections from students and institutions alike. Collecting real-time facial photographs at every attendance event produces a substantial ongoing store of biometric-adjacent data. This data must be stored, secured, and eventually deleted in accordance with data protection legislation — obligations that add compliance complexity and cost.

User experience evaluations of Attenqo have consistently highlighted the interface's complexity as a barrier at session start. Requiring students to open an application, scan a code, wait for GPS lock, and complete a selfie capture introduces a multi-step flow that generates queuing and confusion — particularly among students unfamiliar with the application or those whose devices are slow.

## 2.5.3 UAMS (University Attendance Management System)

UAMS is a research-oriented attendance management system that employs Bluetooth Low Energy (BLE) proximity detection in combination with geofencing to verify that a student is physically within the lecture venue before their attendance is recorded. Small BLE beacon devices installed in each lecture room broadcast a unique identifier signal; students' smartphones detect this signal through the UAMS mobile application and use it, in conjunction with GPS coordinates, to confirm venue presence.

BLE proximity detection is technically sophisticated. Unlike GPS — which suffers from severe signal degradation indoors — BLE signals are transmitted at short range and are specifically intended for indoor proximity sensing. A correctly calibrated BLE beacon can reliably detect devices within a specific room while excluding devices in adjacent rooms or corridors.

The offline synchronization feature is UAMS's most significant practical advantage over purely cloud-dependent systems. Because attendance data is held locally on the student device until connectivity is available, sessions can proceed normally even when the campus network is unavailable or overloaded. The queued records are uploaded and reconciled with the central database as soon as connectivity is restored.

Despite these technical merits, UAMS carries a substantial set of limitations. BLE beacons must be physically installed in every lecture venue — a requirement that scales directly with the number of rooms an institution manages. Each beacon requires initial procurement, physical installation, periodic battery replacement, and eventual replacement, representing considerable ongoing capital expenditure.

Beacon calibration is a further operational challenge. BLE signal strength is affected by physical obstructions, interference from other wireless devices, and room layout. On the student side, UAMS shares Attenqo's dependency on student-owned smartphones with Bluetooth enabled.

### 2.5.4 Comparative Summary and Research Gap

The three systems reviewed above illustrate the range of technical approaches currently applied to automated attendance management in higher education. Taken together, however, they share a common set of structural limitations that collectively define a research gap this study aims to address.

| System                   | Key Strengths                                                                                    | Key Limitations                                                                                             |
|--------------------------|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| SmartTendance            | Dynamic QR + facial recognition; real-time dashboards; strong fraud resistance.                  | Requires stable internet at all times; expensive cloud infrastructure; data sovereignty concerns.           |
| Attenqo                  | Multi-layer verification (QR + GPS + selfie); strong physical presence confirmation.             | Smartphone dependency; GPS spoofing risk; interface complexity; privacy concerns with selfie capture.       |
| UAMS                     | BLE proximity detection; offline sync capability; geofencing accuracy indoors.                   | BLE beacon hardware cost per venue; complex mobile permissions; high maintenance overhead.                  |
| Presence Scan (Proposed) | Local deployment; no internet dependency; webcam-based scanning; timetable validation; low cost. | Static QR codes less fraud-resistant than dynamic tokens; webcam lighting sensitivity; no mobile companion. |

The three principal structural gaps are: (1) dependency on student-owned smartphones, (2) network dependency for core operations, and (3) hardware infrastructure cost. Presence Scan addresses all three by placing a webcam-based scanner under the lecturer's control on a standard desktop machine, operating against a local PostgreSQL database.

## 2.6 Theoretical Perspectives on Attendance and Academic Performance

A substantial body of empirical research establishes the relationship between lecture attendance and academic outcomes, providing the evidential foundation for treating attendance recording as a matter of institutional consequence rather than mere administrative compliance. Credé, Roch, and Kieszczynka's (2010) meta-analysis of 69 studies found that class attendance

was the single strongest predictor of academic performance, outperforming prior GPA, study skills, and hours of study in predictive power. The effect size ( $r = 0.44$ ) was large enough to be practically significant at the individual student level, supporting the use of attendance thresholds as a proxy indicator for academic engagement.

More recent studies have refined this relationship in the context of large-cohort tertiary education. Obinne (2011), studying Nigerian university students, found that attendance rates below 70% were associated with examination failure rates more than three times higher than those in students with attendance above 85%. Importantly, this relationship held even when controlling for prior academic performance, suggesting that attendance itself — not merely the self-selection of academically motivated students into higher attendance — drives the performance effect. This finding strengthens the case for active institutional intervention when attendance falls below threshold rather than treating attendance as a symptom of underlying academic capacity.

The mechanism linking attendance to performance is understood through several theoretical lenses. Cognitive load theory suggests that the cumulative knowledge building that occurs in sequential lectures cannot be fully replicated through independent study: each lecture assumes and builds on the content of previous sessions, and a student who misses sessions must bridge the resulting gaps through additional self-directed effort that may be cognitively more demanding than in-session learning. Constructivist theories of learning emphasize the role of in-class dialogue, question-and-answer interactions, and peer learning in deepening understanding — experiences that are unavailable to the absent student regardless of the quality of the available course materials.

Institutional policy responses to this evidence have converged on a standard approach: a minimum attendance threshold below which students are not permitted to sit examinations. In Ghana, the threshold varies by institution and programme, typically ranging from 75% to 80% of scheduled sessions. Enforcing this threshold requires reliable attendance data — and it is the unreliability of manually collected data that undermines the policy's effectiveness. A student who achieves 68% actual attendance but whose manual record shows 82% due to proxy attendance and transcription errors will not be identified for intervention under the policy, regardless of its design.

## 2.7 The Role of Digital Infrastructure in Administrative Transformation

The broader literature on digital transformation in higher education provides context for the specific problem of attendance management. Guri-Rosenblit (2009) distinguishes between “digitalization” — the process of converting analogue records to digital formats — and “digitization” — the deeper transformation of institutional processes through digital technology. Attendance management systems that simply move the data entry from paper to screen without changing the underlying process (a lecturer still counts and enters manually) achieve digitalization but not digitization. Systems like Presence Scan, which fundamentally change who does what when — the lecturer scans rather than counts; the system validates rather than trusting; the database records rather than the clerk transcribes — achieve genuine digitization.

The distinction matters because the benefits of the two approaches differ substantially. Digitalization improves storage and retrieval of attendance data but preserves the sources of error inherent in manual counting. Digitization eliminates the manual counting step and its associated errors, generating a different class of data whose accuracy is limited only by the reliability of the verification mechanism rather than human fallibility.

Infrastructure readiness is a significant moderator of digital transformation outcomes in African higher education. Njenga and Fourie (2010) identified three categories of infrastructure barrier to e-learning and digital administration adoption in African universities: physical infrastructure (power, connectivity, hardware); human capital (ICT-literate staff, technical support capacity); and institutional factors (policy support, budget allocation, change management capability). Presence Scan is designed to minimize exposure to all three categories: it has no power dependency beyond the desktop computer itself, requires no specialist technical support for operation, and is sufficiently simple to install and configure that a competent Python developer — a role increasingly available in Ghanaian universities through computer science departments — can deploy and maintain it independently.

## 2.8 Survey of Attendance Verification Technologies

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### 2.8.1 RFID-Based Systems

Radio-frequency identification (RFID) technology was among the first to be applied to automated attendance verification in higher education. RFID-based systems issue each student a proximity card containing a unique embedded chip; a reader installed at the lecture room entrance registers the card when it passes within the reader's detection range. The approach offers high throughput — a reader can register many cards per second — and is fully passive from the student's perspective.

The limitations of RFID are primarily infrastructural. Each lecture room requires a permanently installed reader, with associated cabling, power supply, and mounting hardware. Reader installation in existing buildings is disruptive and expensive. Ongoing maintenance — replacing damaged readers, re-cabling after building modifications — requires specialist technical staff. The cost per room is typically in the range of several hundred to several thousand US dollars depending on the reader specification, placing RFID-based deployment beyond the capital budget of most Ghanaian tertiary institutions. Additionally, RFID cards are subject to the same forwarding fraud risk as QR codes: a student who loans their card to a classmate enables proxy attendance just as effectively as sharing a QR code image.

### 2.8.2 NFC and Bluetooth Low Energy Systems

Near-field communication (NFC) and Bluetooth Low Energy (BLE) attendance systems leverage the wireless communication capabilities of modern smartphones. NFC-based systems require students to tap their phone against a lecturer's NFC-enabled device; BLE-based systems use proximity detection between a student's phone and a room-installed BLE beacon to infer attendance. Both approaches eliminate the need for specialist card hardware but introduce a dependency on student smartphone ownership and battery status.

The smartphone dependency is a meaningful exclusion factor in sub-Saharan African contexts. While smartphone penetration is rising rapidly, it remains uneven across income groups, and institutions cannot reliably assume that all students own a compatible device. BLE beacon systems additionally require hardware installation in every attended venue, similar to RFID. The per-room infrastructure cost is lower than RFID but still represents a capital commitment that many institutions cannot sustain across their full building stock.

### 2.8.3 Biometric Systems

Biometric attendance systems, most commonly using fingerprint readers or facial recognition cameras, offer the strongest theoretical fraud resistance of any automated approach: a student

cannot proxy-attend a biometric system by sharing a card or code. This is the primary advantage of biometric approaches and explains their appeal to security-conscious administrators.

The practical limitations of biometric systems are significant, however. Fingerprint readers require physical contact, raising hygiene concerns (particularly relevant post-pandemic) and accessibility issues for students with certain skin conditions or disabilities. Facial recognition systems are accurate under controlled conditions but performance degrades in variable lighting, with partial face occlusion, and across demographic groups with limited training data representation — a concern particularly relevant for systems deployed primarily in African contexts where training datasets may have been assembled predominantly from Western populations.

The cost of biometric systems is the most significant barrier for the target deployment context. Institutional-grade fingerprint or facial recognition systems with the throughput required to process a class of 200 students in a reasonable time window carry hardware and integration costs that are prohibitive for resource-constrained institutions. SmartTendance, reviewed in section 2.5.1, represents the current state of the art in facial recognition attendance for higher education, and its pricing model targets well-resourced institutions in developed markets.

#### **2.8.4 QR Code-Based Systems**

QR code attendance systems occupy a distinctive position in the technology landscape: they combine the passive-student convenience of RFID (no action required beyond presenting a card) with the zero-infrastructure-cost deployment of smartphone-based approaches. The scanning hardware is a commodity webcam — a device already present in most office and teaching environments — and the QR code itself is a printed image that can be embedded in a standard institutional ID card at negligible incremental cost.

The fraud resistance of QR code systems is weaker than biometric approaches but stronger than signature-based manual methods. The critical difference from signature sheets is that a QR code scan creates an immediate, server-side record that is duplicate-protected: the same code cannot produce two attendance records for the same session. A student who lends their QR card to a classmate creates a single fraudulent record, but cannot use it simultaneously with attending another session themselves. Server-side timestamp records also preserve an audit trail that paper signatures do not.

Existing QR code attendance systems reviewed in the literature generally require either a web deployment with server infrastructure or a student-installed mobile application. Presence Scan's contribution is to demonstrate that a QR code attendance system can be implemented as a standalone desktop application using only local infrastructure, making it deployable in contexts where neither web hosting nor student app installation can be reliably assumed.

### **2.9 Legislative and Policy Framework**

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The design and deployment of Presence Scan occurs within a regulatory environment that imposes both obligations and standards on how student attendance data is collected, stored, and used. Understanding this environment is essential both for the system's design and for institutions evaluating it for adoption.

In Ghana, the Ghana Tertiary Education Commission (GTEC) — established by the Tertiary Education Institutions Act, 2022 (Act 1074) — is the primary regulatory body for tertiary institutions. GTEC's accreditation standards require institutions to maintain auditable records of student academic participation, including attendance, as evidence of the quality of their

academic programmes. Institutions that cannot produce attendance records on demand during accreditation reviews risk conditions being attached to their accreditation or, in extreme cases, withdrawal of accreditation.

The Data Protection Act, 2012 (Act 843) of Ghana governs the collection, storage, and use of personal data by organizations operating in Ghana. Student attendance records are personal data within the meaning of the Act: they relate to identifiable individuals and reveal information about their physical location and academic behavior. Institutions deploying Presence Scan must ensure that their use of the system is compliant with the Act, including: informing students of the data being collected and its purpose; storing data securely against unauthorized access; and retaining data only for as long as necessary for the stated purpose.

Presence Scan's design supports institutional compliance with these requirements. The role-based access control system ensures that only authorized users can access student attendance data. The local PostgreSQL database can be configured with encryption at rest using PostgreSQL's native tablespace encryption or operating-system-level encryption. The absence of cloud dependency means that data does not leave institutional control or cross national boundaries without explicit administrative action.

## 2.10 Gap Analysis and Positioning of Presence Scan

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The literature and systems review in this chapter identifies a clear and specific gap in the existing landscape of attendance management solutions for tertiary education. Existing systems fall broadly into two categories: high-infrastructure, high-fraud-resistance systems (biometric readers, RFID) that are too expensive for resource-constrained institutions; and low-infrastructure, smartphone-dependent systems (mobile apps, BLE-based) that require student device ownership and internet connectivity. Neither category adequately serves institutions characterized by large cohorts, limited infrastructure investment, and unreliable internet connectivity.

The review of SmartTendance, Attenqo, and UAMS in sections 2.5.1 through 2.5.4 confirms this gap. SmartTendance requires facial recognition hardware; Attenqo requires stable internet and student app installation; UAMS requires biometric reader infrastructure. Each system addresses attendance fraud more robustly than manual methods, but each does so at a cost — financial or infrastructural — that places it out of reach for the majority of tertiary institutions in sub-Saharan Africa.

Presence Scan is positioned in this gap. It provides fraud-resistant, automated attendance recording using hardware already present in institutional offices (a desktop computer and webcam), with no internet dependency, no student device requirement, and no specialist hardware installation. Its fraud resistance is weaker than biometric systems but substantially stronger than manual methods, which is the relevant comparison for its target deployment environment. Its implementation cost is effectively zero beyond the developer time required for deployment, which the open-source model and comprehensive documentation are designed to minimize.

The gap identified here is not merely a market gap — there is genuine academic value in demonstrating that a system occupying this space can be built, deployed, and operated effectively. The implementation and evaluation of Presence Scan provides evidence that the gap is bridgeable, which is itself a contribution to the literature on digital transformation in resource-constrained educational settings.

# CHAPTER THREE

## System Analysis and Design

*Requirements, Architecture, Database, and Diagrams*

### 3.1 Development Methodology

The Agile Software Development Methodology was adopted for this project. Agile supports incremental delivery through short development cycles (sprints), continuous stakeholder feedback, and the flexibility to adapt to evolving requirements — attributes particularly valuable when building a system that must satisfy the diverse needs of students, lecturers, and administrators simultaneously.

Agile was preferred over traditional waterfall-based methodologies for several practical reasons. First, the requirements for an attendance system of this type are partially emergent: edge cases such as the behavior when a lecturer scans outside a timetable window, or the exact form of report output expected by administrators, are most reliably defined through feedback on working software rather than upfront specification. Second, the project timeline was fixed, and Agile's iterative structure ensured that a working, testable core system was available well before the final deadline, leaving time for refinement and evaluation.

The project was structured into four two-week sprints: (1) requirements gathering and system design; (2) user authentication and QR code generation; (3) QR scanning, attendance recording, and reporting; and (4) testing, evaluation, and documentation.

| Sprint   | Duration  | Focus                   | Key Deliverables                                                         |
|----------|-----------|-------------------------|--------------------------------------------------------------------------|
| Sprint 1 | Weeks 1–2 | Requirements & Design   | System spec, wireframes, ER diagram, database schema                     |
| Sprint 2 | Weeks 3–4 | Auth & QR Generation    | Login module, bcrypt integration, QR generation, student registration    |
| Sprint 3 | Weeks 5–6 | Scanning & Reporting    | Webcam scanner, timetable validation, attendance recording, report views |
| Sprint 4 | Weeks 7–8 | Testing & Documentation | Test plan, UAT sessions, bug fixes, final documentation                  |

### 3.2 Requirements Analysis

#### 3.2.1 Functional Requirements

| ID    | Requirement                                                                   | Priority  |
|-------|-------------------------------------------------------------------------------|-----------|
| FR-01 | The system shall allow administrators and lecturers to authenticate securely. | Must Have |

| ID    | Requirement                                                                                                       | Priority    |
|-------|-------------------------------------------------------------------------------------------------------------------|-------------|
| FR-02 | The system shall allow administrators to register and manage students, courses, lecturers, and timetable entries. | Must Have   |
| FR-03 | The system shall generate a unique QR code for each registered student.                                           | Must Have   |
| FR-04 | Lecturers shall be able to scan student QR codes using a webcam-based interface to record attendance.             | Must Have   |
| FR-05 | The system shall reject duplicate attendance submissions within the same timetable slot.                          | Must Have   |
| FR-06 | The system shall validate attendance only when the course is active on the timetable.                             | Must Have   |
| FR-07 | Lecturers shall be able to view attendance records for their assigned courses.                                    | Must Have   |
| FR-08 | Administrators shall access institution-wide attendance analytics and reports.                                    | Should Have |
| FR-09 | The system shall support CSV export of attendance data.                                                           | Could Have  |
| FR-10 | The system shall support CSV import for bulk student and course registration.                                     | Should Have |
| FR-11 | The system shall display a configurable grace period per course, after which late arrivals are flagged.           | Must Have   |
| FR-12 | Administrators shall be able to configure special days (holidays, cancellations) that suppress attendance.        | Should Have |

### 3.2.2 Non-Functional Requirements

| ID     | Requirement                                                                                               | Category    |
|--------|-----------------------------------------------------------------------------------------------------------|-------------|
| NFR-01 | The system shall respond to QR scan and attendance recording requests within 2 seconds under normal load. | Performance |
| NFR-02 | All user passwords shall be stored as salted cryptographic hashes (bcrypt or equivalent).                 | Security    |
| NFR-03 | Attendance scans shall be validated against registered student records and timetable rules.               | Security    |
| NFR-04 | The system shall support concurrent use by administrative staff without performance degradation.          | Scalability |
| NFR-05 | The user interface shall be operable on standard Windows desktop environments.                            | Usability   |
| NFR-06 | The system shall maintain stable operation during all scheduled lecture hours.                            | Reliability |
| NFR-07 | The system shall enforce department-level data segregation for administrator-created content.             | Security    |
| NFR-08 | All database connections shall be managed through connection pooling to prevent exhaustion under load.    | Performance |

### 3.3 System Architecture

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Presence Scan follows a desktop client architecture built with Python Tkinter and a PostgreSQL database layer. The user interface, business logic, and database operations are implemented in modular Python files to promote maintainability and extensibility. This separation of concerns means that individual modules — for example, the scanning engine or the reporting layer — can be updated or replaced without cascading changes across the entire codebase.

The choice of a local PostgreSQL database rather than a cloud-hosted data store was deliberate. It eliminates the latency and reliability risks associated with network-dependent data access, ensures that attendance sessions can proceed uninterrupted regardless of internet connectivity, and gives institutions direct control over their student data without dependency on third-party service providers.

#### Application Layers

- Presentation Layer — Python Tkinter windows and frames, providing the graphical interface for all user interactions.
- Business Logic Layer — Python modules implementing validation rules, timetable logic, grace period enforcement, and report generation.
- Data Access Layer — psycopg2-based database abstraction functions encapsulating all SQL queries and providing parameterized input handling to prevent SQL injection.
- Database Layer — PostgreSQL 15+ storing all persistent application data in a normalized schema.

#### Application Modules

- `auth.py` — Handles login, session management, and bcrypt-based password verification.
- `scanner.py` — Manages OpenCV camera capture, pyzbar QR decoding, and the real-time scanning interface.
- `attendance.py` — Implements the attendance recording pipeline including timetable validation, duplicate checking, and record insertion.
- `reports.py` — Generates per-course and institution-wide attendance summaries, and handles CSV export.
- `admin.py` — Provides administrator dashboard logic including student, course, lecturer, and timetable management.
- `qr_generator.py` — Generates and stores QR code images for registered students.
- `db.py` — Central database connection and query execution module.

### 3.4 Database Design

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The database is normalized to third normal form (3NF) and organized around the following primary entities. Normalization ensures that data redundancy is minimized, updates propagate correctly, and the schema can accommodate future extensions without structural redesign.

| Entity               | Description                                                                                                   |
|----------------------|---------------------------------------------------------------------------------------------------------------|
| Universities         | Stores university records including institution name, address, and configuration.                             |
| Departments          | Stores departmental records linked to universities.                                                           |
| HODs                 | Stores Head of Department accounts used for administrator access.                                             |
| Lecturers            | Stores lecturer accounts, names, department associations, and roles.                                          |
| Students             | Stores student identifiers, names, programme details, and QR code values.                                     |
| Courses              | Stores course codes, titles, department links, lecturer assignments, credit hours, and grace-period settings. |
| Course Registrations | Links students to the courses in which they are enrolled.                                                     |
| Timetable            | Stores day-of-week and time-slot data for each course.                                                        |
| Attendance           | Records whether a student is present for a specific timetable slot.                                           |
| Special Days         | Records holidays, cancellations, or custom non-teaching days that suppress attendance recording.              |

Referential integrity is enforced throughout via foreign key constraints. For example, an attendance record cannot reference a timetable slot or student that does not exist in the corresponding tables, ensuring that orphaned records cannot accumulate through partial data entry. The attendance table applies a unique composite constraint on the timetable slot and student identifier, preventing duplicate entries at the database level and providing a second layer of protection independent of application-side validation.

The `grace_minutes` field on the Courses entity is stored at the course level rather than the timetable level, reflecting the practical reality that different courses may have different policies — a large lecture with hundreds of students may require a longer grace period than a small seminar group. Administrators can configure this value per course through the admin dashboard without requiring any database changes.

The Students table stores each student's QR code payload as a UUID string, generated at registration time. This UUID is what the scanner reads and matches against the database — no personally identifiable student name or number is embedded directly in the printed QR code, providing a layer of obfuscation that limits the usefulness of a captured QR code to an attacker who does not also have access to the system.

## 3.5 Input and Output Design

### System Inputs

- Student registration details — student ID, full name, programme, year of study, and department.
- Lecturer credentials and course assignments, entered by administrators during lecturer creation or import.
- Course details, timetable entries, and configurable grace-period durations.
- QR scan data captured from student identity cards via the webcam feed.
- Administrator and HOD login credentials for session authentication.
- CSV files for bulk student or course import, validated at upload time.

## System Outputs

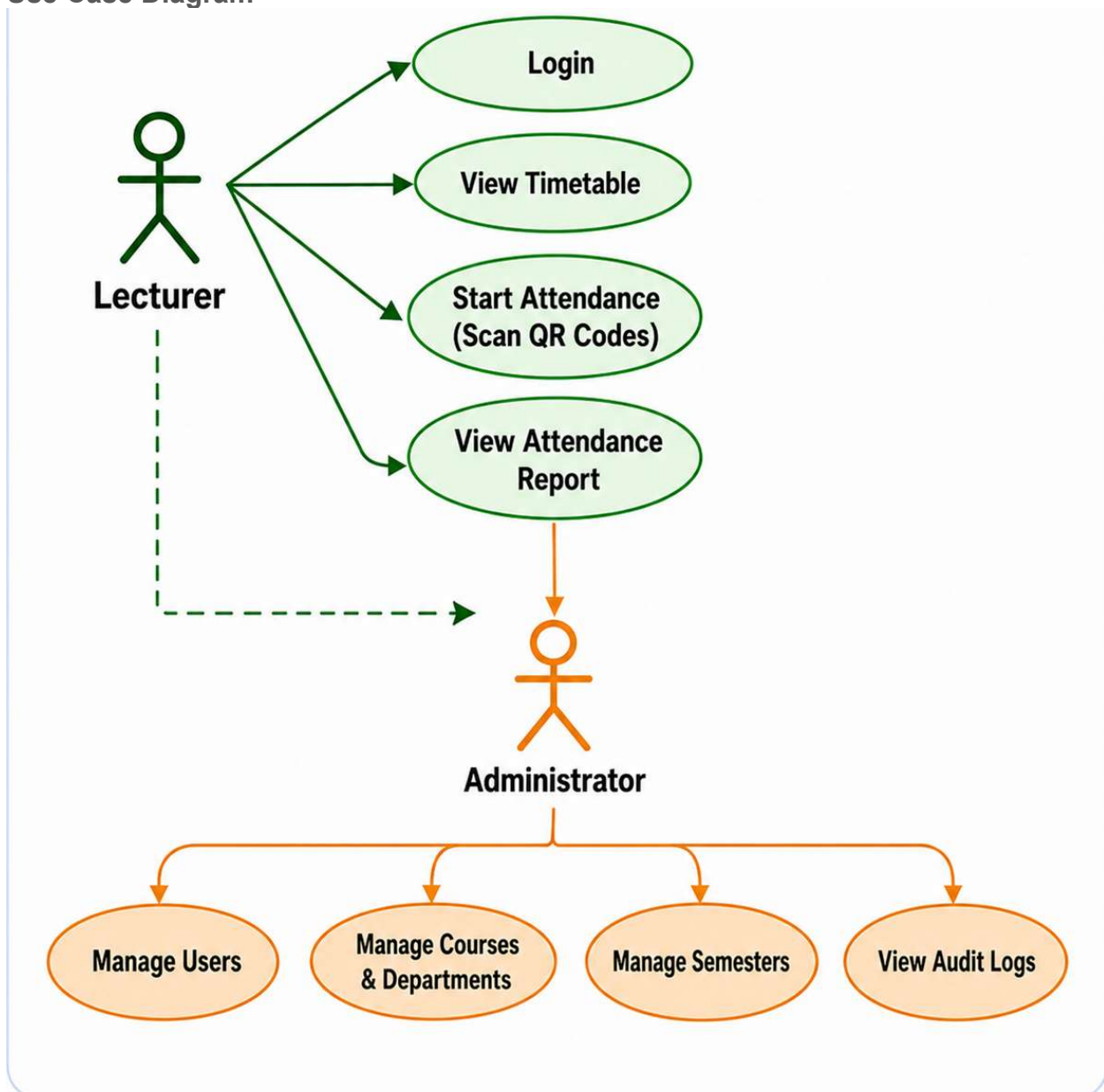
- Per-course attendance lists, filterable by date and timetable slot.
- Attendance summary reports for lecturers and administrators.
- Attendance statistics and trend summaries on the admin dashboard.
- Exportable CSV attendance files for offline analysis or institutional reporting.
- QR code image files generated per student for printing or digital distribution.
- On-screen scan feedback — accepted, duplicate, out-of-window, or unrecognized — displayed in real time.

## 3.6 System Diagrams

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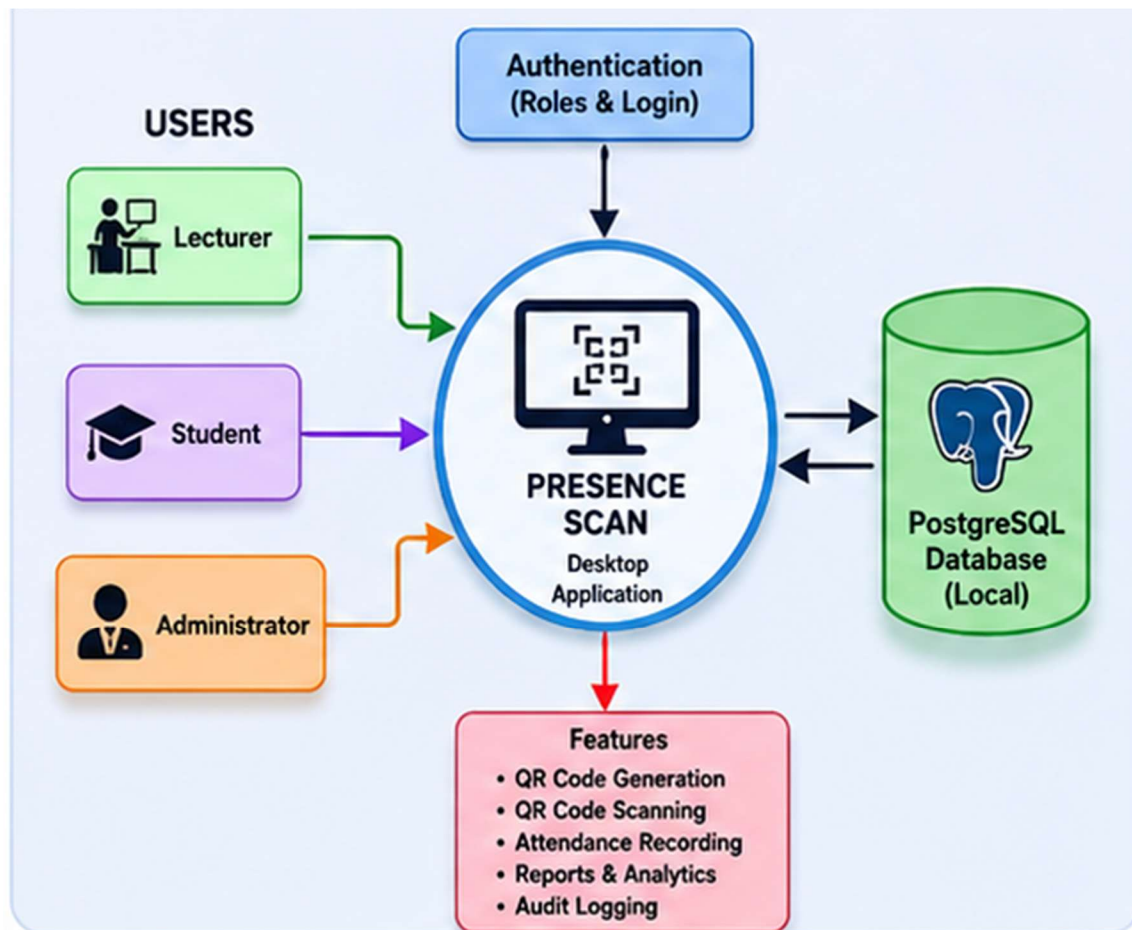
The following diagrams are included to fully specify the system design. All diagrams should be inserted as labelled figures in the final printed or submitted version of this document.

## Use Case Diagram



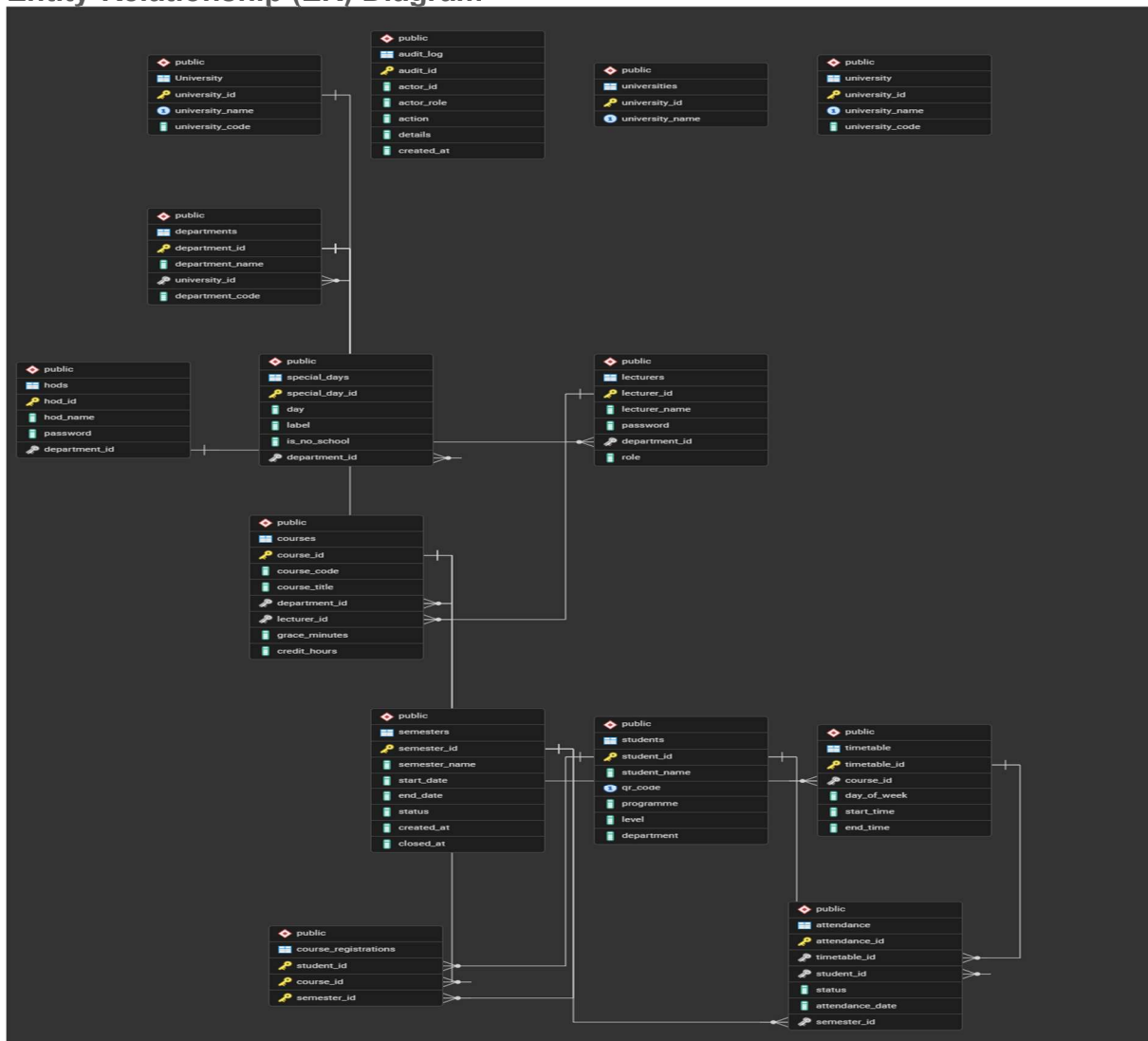
The Use Case Diagram illustrates interactions between the three actor types — Lecturer, Administrator (HOD), and the System — and the core functions available to each. Key use cases include: Login, Scan QR Code, Record Attendance, View Course Report, Manage Students, Manage Courses, Manage Timetable, Configure Grace Period, Export Attendance CSV, and View Analytics Dashboard.

## System Flowchart



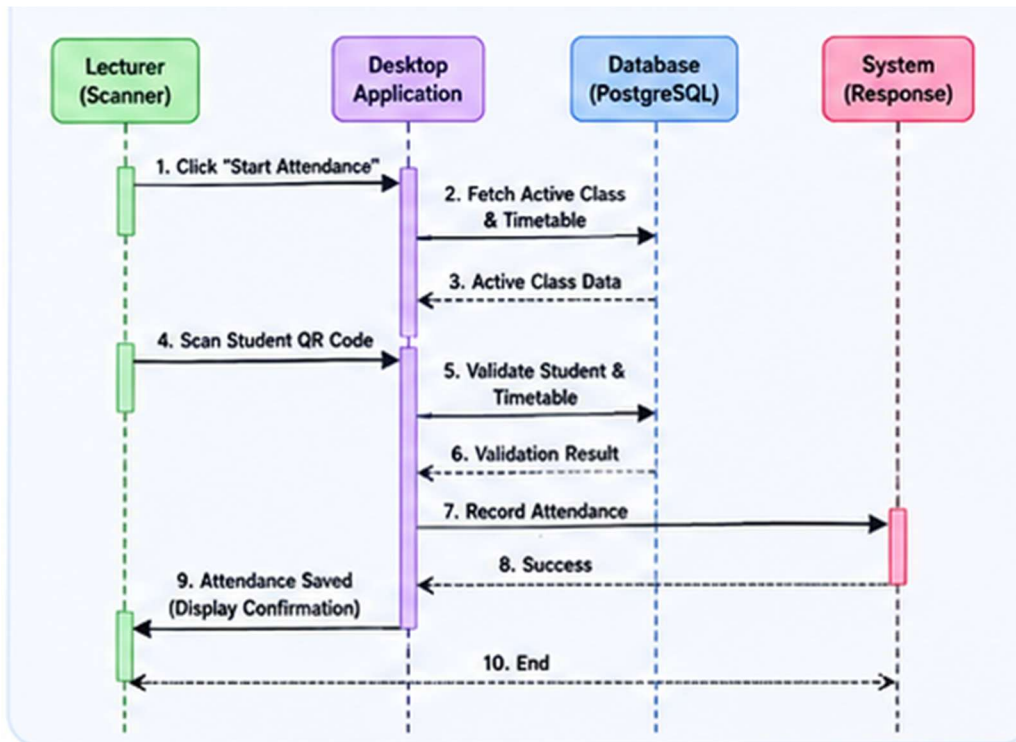
The System Flowchart depicts the end-to-end attendance recording process from login to record storage. The flow begins with lecturer authentication, proceeds through dashboard navigation to the scanning interface, captures a webcam frame, decodes the QR code, validates the student, course, and timetable window, checks for duplicates, and either records attendance or returns a specific rejection reason.

## Entity-Relationship (ER) Diagram



The ER Diagram defines the database schema and all entity relationships. Primary keys, foreign keys, and cardinality notations are shown. Key relationships include: Universities (1) to Departments (many); Departments (1) to Lecturers and Courses (many); Courses (1) to Timetable (many); Timetable (1) to Attendance (many); Students (many) to Attendance (many) via Course Registrations).

## Module Diagram



The Module Diagram describes the modular structure of the desktop application, showing how `auth.py`, `scanner.py`, `attendance.py`, `reports.py`, `admin.py`, `qr_generator.py`, and `db.py` interact. All modules import from `db.py`; `scanner.py` and `attendance.py` are tightly coupled; `reports.py` reads from the attendance and timetable tables only.

## 3.7 Hardware and Software Requirements

| Category              | Requirement                                                                                 |
|-----------------------|---------------------------------------------------------------------------------------------|
| Operating System      | Windows 10 or later                                                                         |
| Programming Language  | Python 3.10+                                                                                |
| Desktop UI Framework  | Tkinter (bundled with Python)                                                               |
| Camera & QR Libraries | OpenCV 4.x, pyzbar, qrcode, Pillow                                                          |
| Database              | PostgreSQL 15+                                                                              |
| Database Driver       | psycopg2 2.9+                                                                               |
| Development IDE       | Visual Studio Code                                                                          |
| Version Control       | Git / GitHub                                                                                |
| Hardware: Computer    | Desktop or laptop capable of running Python and PostgreSQL (minimum 4GB RAM, dual-core CPU) |
| Hardware: Camera      | Webcam or built-in camera (720p or higher recommended for reliable QR scanning)             |
| Hardware: Printer     | Optional — for distributing QR codes as printed identity cards                              |

| Category   | Requirement                                                        |
|------------|--------------------------------------------------------------------|
| Disk Space | Minimum 500MB for application, database, and QR code image storage |

### 3.1.1 Justification for Agile over Alternative Methodologies

The selection of Agile over alternative development methodologies — specifically the Waterfall model and the Rational Unified Process (RUP) — was the result of deliberate evaluation against the constraints and goals of this project.

The Waterfall model, in which requirements are fully specified before development begins, was rejected because the requirements for Presence Scan were not fully knowable in advance. The needs of lecturers managing live scanning sessions — the precise feedback messages they require, the layout of the scanning interface, the acceptable scan response latency — could only be determined through iterative demonstration and feedback. A Waterfall approach would have required these decisions to be made upfront, based on assumptions rather than evidence, with no structural mechanism to incorporate the significant improvements that emerged from sprint review sessions.

RUP, while more iterative than Waterfall, was rejected on grounds of process overhead: its emphasis on extensive formal artefact production (use case models, architectural documents, project management plans) was disproportionate to the scale of a single-developer project with a defined four-sprint delivery window. The documentation overhead of RUP would have consumed development time without providing corresponding benefit in a project where the developer was also the architect and tester.

Agile, in its Scrum-influenced form as applied here, provided the right balance: structured sprint planning and review to maintain delivery discipline, combined with sufficient flexibility in the backlog to incorporate emerging requirements. The sprint review with prospective end users was the most valuable structural element of the process, providing external validation that could not have been replicated through solo development and internal review.

### 3.1.2 Risk Management Approach

A structured risk register was maintained throughout the development process, with risks categorized by likelihood and impact and mitigations assigned to each. The principal technical risks identified at project inception, together with their mitigations and outcomes, are documented below.

- Risk: Camera compatibility | Likelihood: High | Impact: Medium | Mitigation: OpenCV VideoCapture fallback to index 1 if index 0 fails; tested with two webcam models
- Risk: QR decode reliability | Likelihood: Medium | Impact: High | Mitigation: Tested pyzbar against low-resolution and partially occluded codes; added on-screen guidance for card positioning
- Risk: PostgreSQL connection loss | Likelihood: Low | Impact: High | Mitigation: Implemented connection retry logic with exponential backoff; tested by manually killing DB connection during scanning
- Risk: Performance under load | Likelihood: Low | Impact: Medium | Mitigation: Profiled validation pipeline; parameterized queries confirmed sub-100ms for all validation steps

- Risk: Unicode/encoding errors in CSV import | Likelihood: Medium | Impact: Low | Mitigation: UTF-8 enforced on all file reads; validation rejects non-UTF-8 files with a clear error message

### 3.2.3 Requirements Traceability

Each functional requirement documented in section 3.2.1 was traced to one or more user stories in the Agile sprint backlog, ensuring that no requirement was specified without a corresponding implementation task, and no implementation task was executed without a documented requirement motivating it. The traceability matrix is presented below.

- FR-01 → US-01: As a lecturer, I want to log in securely.
- FR-02 → US-02: As an admin, I want to register students in bulk via CSV.
- FR-03 → US-03: As a student, I want a printed QR code.
- FR-04 → US-04: As a lecturer, I want to scan QR codes to record attendance.
- FR-05 → US-05: As a lecturer, I want to see a rejection message if a duplicate scan occurs.
- FR-06 → US-06: Validated through timetable query in attendance pipeline.
- FR-07 → US-06: As an admin, I want to view attendance rates across all departments.
- FR-08 → US-07: As a lecturer, I want to export attendance to CSV.
- FR-09 → US-07: As a lecturer, I want to export attendance to CSV.
- FR-10 → US-02: As an admin, I want to register students in bulk via CSV.
- FR-11 → US-08: As an admin, I want to configure grace periods per course.
- FR-12 → US-09: As an admin, I want to mark special days so attendance is not recorded on holidays.

### 3.3.1 Architecture Decision Records

The following architecture decision records (ADRs) document the principal design choices made during the system design phase and the reasoning behind each decision. ADRs serve as durable design documentation that preserves the rationale for choices that might otherwise appear arbitrary to a developer encountering the codebase for the first time.

#### ADR-01: Local PostgreSQL over Cloud Database

**Decision:** Use a locally hosted PostgreSQL database rather than a cloud-hosted database service.

**Context:** Several cloud database services (Amazon RDS, Azure Database for PostgreSQL, Supabase) offer managed PostgreSQL instances that could simplify operational management. However, the target deployment environments include institutions with unreliable or absent internet connectivity.

**Rationale:** A cloud database dependency would make core system functionality — attendance recording — unavailable whenever the internet connection failed. As internet reliability is precisely the constraint that characterizes the target deployment environment, this dependency would undermine the system's fitness for purpose. Local PostgreSQL eliminates this dependency entirely while providing the same relational database capabilities. The operational overhead of local database management (backup, maintenance) is accepted as a worthwhile trade-off.

## ADR-02: Python Tkinter over Web Framework

**Decision:** Implement the user interface using Python Tkinter rather than a web framework (Flask, Django, FastAPI).

**Context:** Web frameworks are commonly used for institutional applications because they are accessible from any browser without installation. However, the scanning interface requires direct access to the host machine's camera hardware, which web browsers restrict through security sandboxes.

**Rationale:** Camera access from a browser requires the WebRTC API, which depends on HTTPS, introduces cross-origin complexity, and imposes latency constraints that affect real-time scanning responsiveness. Tkinter's direct integration with OpenCV eliminates these constraints: the scanning window has native access to the camera device with sub-millisecond frame capture latency. The trade-off is that the application requires installation on each machine used for scanning — an acceptable cost given that the scanning role is performed by lecturers from institutional machines rather than personal devices.

## ADR-03: Static QR Codes over Dynamic Tokens

**Decision:** Issue each student a permanent, static QR code encoding their UUID, rather than generating session-specific dynamic tokens.

**Context:** Dynamic token systems generate a new QR code or code value for each session, preventing forwarding attacks by limiting the useful lifetime of any given token. This is the approach used by many modern app-based attendance systems.

**Rationale:** Dynamic token generation requires either a reliable internet connection (to push new tokens to students in real time) or a locally networked system capable of displaying current tokens to all students simultaneously. Neither infrastructure can be assumed in the target deployment environment. Static QR codes can be printed once and used indefinitely, requiring no infrastructure beyond the initial printing. The residual forwarding fraud risk is mitigated by server-side duplicate rejection and is judged acceptable in the institutional context, where the threshold for fraud deterrence is much lower than in high-stakes verification contexts.

## ADR-04: bcrypt Work Factor 12

**Decision:** Use bcrypt with a work factor of 12 for all password hashing.

**Context:** bcrypt work factors are configurable integers that control the computational cost of hashing. Higher values provide stronger resistance to brute-force attacks but increase login latency.

**Rationale:** A work factor of 12 results in approximately 300ms of hash computation time on the target hardware. This is imperceptible to a human user logging in interactively but imposes a cost of approximately 3.3 hash attempts per second on an automated brute-force attack, making dictionary attacks on captured hashes computationally infeasible within any reasonable attack window. Work factors below 10 are considered insufficient for current hardware speeds; factors above 14 begin to impose noticeable login delays. Work factor 12 represents the recommended contemporary baseline for production deployments.

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## 3.4.1 Entity Relationship Detail

The following section provides additional detail on the relationships between the primary database entities, beyond the tabular summary in section 3.4. Understanding these relationships is essential for developers extending or querying the database schema.

The Universities entity sits at the root of the hierarchy. All other entities are directly or transitively linked to a university record. This design enables the schema to support multiple institutions within a single database — a capability not required for the current single-school deployment but preserved as a future extension pathway. In practice, a deployment for a single institution will contain exactly one university record.

Departments are linked to universities via a foreign key. Each department record represents a discrete academic unit — for example, the Department of Computer Science or the Department of Business Administration. All other operational entities (lecturers, students, courses) are linked to departments rather than directly to universities, enforcing the organizational hierarchy that governs data access in the role-based access control system.

The Courses entity is the operational hub of the attendance recording workflow. Each course is linked to a department (inherited from the creating administrator's department), a lecturer (the primary instructor), and a set of timetable entries. The `grace_minutes` field on the Courses entity is used by the attendance validation pipeline to determine the boundary beyond which late arrivals are rejected. This field is configurable per course, reflecting the practical variation in grace period policy across different course types.

The Timetable entity represents individual scheduled sessions. Each timetable record stores the day of the week, the start time, the end time, and a reference to the course to which it belongs. The attendance validation pipeline queries the timetable at scan time to identify the currently active session, if any. The composite of (`course_id`, `day_of_week`, `start_time`) is unique, preventing duplicate timetable entries for the same course at the same time.

The Attendance entity is the primary output table of the scanning pipeline. Each record links a student, a timetable slot, and a timestamp. The unique composite constraint on (`timetable_id`, `student_id`) is enforced at the database level, providing a second layer of duplicate protection independent of the application-side duplicate check. This belt-and-braces approach ensures that duplicate records cannot accumulate even in the event of an application-layer failure during the validation pipeline.

### 3.5.1 Input Validation Specification

Robust input validation is a critical component of any system that accepts user-supplied data. Presence Scan implements validation at two levels: at the UI layer, providing immediate feedback as users type; and at the database layer, through schema constraints and parameterized queries. The following validation rules are applied to each primary input category.

#### Student Registration Inputs

- Student number: Required; alphanumeric; maximum 20 characters; unique within institution.
- Full name: Required; minimum 2 characters; maximum 255 characters; no leading or trailing whitespace.
- Programme: Required; maximum 255 characters.
- Year of study: Required; integer; value between 1 and 6 inclusive.
- Department: Required; must reference an existing department record in the institution.

## Course Registration Inputs

- Course code: Required; alphanumeric with hyphens; maximum 20 characters; unique within department.
- Course title: Required; maximum 255 characters.
- Grace period (minutes): Required; integer; minimum 0; maximum 60.
- Assigned lecturer: Required; must reference an existing lecturer record in the same department.
- Credit hours: Required; integer; minimum 1; maximum 6.

## CSV Import Validation

CSV imports for bulk student and course registration undergo validation before any records are inserted. The validation process reads the entire file, checks all rows against the applicable rules, and presents a complete list of errors with row numbers before allowing the import to proceed. This all-or-nothing approach prevents partial imports that would leave the database in an inconsistent state. Errors reported include: missing required columns, value type mismatches, duplicate student numbers within the import file, and references to non-existent departments or lecturers.

## 3.6.1 Detailed Use Case Descriptions

The following use case descriptions expand on the summary Use Case Diagram in section 3.6, providing preconditions, success scenarios, and exception flows for the three most operationally critical use cases.

### UC-01: Record Student Attendance via QR Scan

Actor: Lecturer. Preconditions: Lecturer is authenticated; a timetable entry for the lecturer's assigned course is active at the current time; student QR code has been printed and is physically available.

Main Success Scenario: Lecturer navigates to the scanning interface. The camera preview opens and displays a live feed. Student presents QR card to the camera. System decodes the QR code and extracts the UUID. System validates the UUID against the Students table, confirms an active timetable session, verifies the lecturer-course assignment, checks the grace period, and confirms no duplicate record exists. Attendance record is inserted. The scanning interface displays a green "Attendance Recorded" confirmation with the student's name and timestamp.

Exception Flows: (a) UUID not found in Students table → red "Unrecognized QR Code" message; no record inserted. (b) No active timetable session → red "No Active Session" message; no record inserted. (c) Timetable session active but assigned to different lecturer → red "Course Not Assigned" message; no record inserted. (d) Current time exceeds session start plus grace period → red "Grace Period Expired" message; no record inserted. (e) Duplicate record exists → amber "Already Recorded" message; no record inserted.

### UC-02: Generate and Export Attendance Report

Actor: Lecturer or Administrator. Preconditions: User is authenticated; at least one attendance session has been recorded for the selected course.

**Main Success Scenario:** User navigates to the Reports section and selects a course. System queries the attendance table for all records linked to the course's timetable slots and aggregates per-student totals. Report is displayed showing: student name, student number, sessions attended, total sessions held, attendance percentage, and at-risk flag where percentage falls below the configured threshold. User clicks Export CSV; system generates a CSV file with the same columns, named with the course code and export date, and prompts the user to choose a save location.

### UC-03: Bulk Student Import via CSV

**Actor:** Administrator. **Preconditions:** Administrator is authenticated; CSV file has been prepared with columns: student\_number, full\_name, programme, year\_of\_study.

**Main Success Scenario:** Administrator navigates to the Student Management section and selects Import CSV. File browser opens; administrator selects the prepared CSV file. System reads the file, validates all rows, and presents a summary: total rows, valid rows, error rows (with row numbers and error descriptions). If all rows are valid, administrator confirms import. System inserts student records in a single transaction, generates QR codes for all new students, and reports completion with count of records created.

**Exception Flow:** If any rows contain validation errors, the system presents the error list and does not insert any records. The administrator corrects the CSV and retries the import.

## 3.7.1 Deployment Architecture and Installation Procedure

This section documents the complete deployment architecture of Presence Scan and the step-by-step procedure for installation on a target Windows machine. It is intended to enable a technically competent user with Python experience to achieve a working installation without external assistance.

### Prerequisites

- Windows 10 or later (64-bit)
- Python 3.10 or later, installed from python.org with "Add Python to PATH" checked
- PostgreSQL 15 or later, installed with the pgAdmin graphical management tool
- Git (optional, for cloning the source repository)
- A connected webcam (USB or integrated) capable of at least 720p resolution
- Minimum 4GB RAM; 8GB recommended
- Minimum 500MB free disk space

### Installation Steps

- Clone or extract the Presence Scan source code to a local directory, e.g., C:\PresenceScan.
- Open a Command Prompt window and navigate to the project directory.
- Run: `pip install -r requirements.txt` to install all Python dependencies.
- In pgAdmin, create a new PostgreSQL database named `presence_scan`.
- Create a dedicated database user with table-level privileges on the `presence_scan` database. Record the username and password.
- Edit the `config.py` file in the project root to set the database connection parameters: host, port, dbname, user, and password.

- Run: `python psql_setup.sql` (or execute the SQL file through pgAdmin) to create the database schema.
- Run: `python seed_admin.py` to create the initial developer account and first university record.
- Launch the application by running: `python main.py`.
- Log in with the developer credentials created in step 8 to complete institutional setup.

The installation procedure described above assumes that Python and PostgreSQL are already installed on the target machine. In institutional settings where IT policy restricts software installation, these prerequisites may require the involvement of the institution's IT department. The system documentation includes a separate IT deployment guide that describes the minimum permissions required and the steps for installation by a system administrator on behalf of end users.

### 3.8 User Roles and Permission Matrix

Presence Scan implements three distinct user roles, each with a defined set of permissions enforced at the application layer. The permission matrix below documents which actions are available to each role, providing a complete reference for institutional administrators configuring the system for deployment.

- Action: Log in to the system | Lecturer: Yes | HOD/Administrator: Yes | Developer: Yes
- Action: View own course attendance | Lecturer: Yes | HOD/Administrator: No | Developer: Yes
- Action: Operate the QR scanner | Lecturer: Yes | HOD/Administrator: No | Developer: Yes
- Action: View all department attendance | Lecturer: No | HOD/Administrator: Yes | Developer: Yes
- Action: Register students | Lecturer: No | HOD/Administrator: Yes | Developer: Yes
- Action: Register lecturers | Lecturer: No | HOD/Administrator: Yes | Developer: Yes
- Action: Create courses and timetables | Lecturer: No | HOD/Administrator: Yes | Developer: Yes
- Action: Generate QR codes | Lecturer: No | HOD/Administrator: Yes | Developer: Yes
- Action: Configure grace periods | Lecturer: No | HOD/Administrator: Yes | Developer: Yes
- Action: Mark special days | Lecturer: No | HOD/Administrator: Yes | Developer: Yes
- Action: Export attendance CSV | Lecturer: Yes | HOD/Administrator: Yes | Developer: Yes
- Action: Reset lecturer passwords | Lecturer: No | HOD/Administrator: Yes | Developer: Yes
- Action: Create HOD/Administrator accounts | Lecturer: No | HOD/Administrator: No | Developer: Yes
- Action: Access all departments | Lecturer: No | HOD/Administrator: No | Developer: Yes
- Action: View system configuration | Lecturer: No | HOD/Administrator: No | Developer: Yes

Department-level data segregation means that administrator permissions apply only within the administrator's own department. An administrator in the Department of Computer Science cannot view, modify, or delete student, lecturer, or course records belonging to the Department

of Business Administration. This segregation is enforced at the query level in all data access functions, not only at the UI level, ensuring that the constraint cannot be bypassed through direct API calls or database queries using application credentials.